

Axiomatizing U and S over Integer Time

M Reynolds¹

Imperial College
LONDON SW7 2AZ.

Abstract. We give a Hilbert style axiomatization for the set of formulas in the temporal language with *Until* and *Since* which are valid over the integer number flow of time. We prove weak completeness for this orthodox axiom system.

1 Introduction

We continue a long tradition of axiomatizing temporal logics. Variations on the theme have been achieved by varying the language used to talk about events in time and by varying the assumptions made about the nature of time. In this paper we consider the particularly interesting case of the language with “until” and “since” over integer number time.

Early axiomatization results were for the language with two temporal connectives: F for “will” and P for “was”. Completeness of axiomatizations for this language over various flows of time can be proved by tinkering with Henkin constructions (see for example [11]).

The stronger, more expressive, language with “until” (U) and “since” (S) presents more difficulties. Burgess [1] gives a more complicated Henkin construction which is sufficient to prove completeness when we consider certain whole classes of flows of time such as linear flows.

Axiomatization over the rational numbers fall out of Burgess’ work easily. However, it is generally more difficult again to get completeness results for specific flows of time. Although axioms for U over the more “useful” natural numbers flow appear with an involved proof in [4] an axiomatization of U and S over the natural numbers flow (along with other classes of well-orderings) has only just been given in [12].

In this paper, we use the techniques developed in [9] to provide an axiomatization for U and S over the integers. After defining the logic in the next section we will show that only a weak completeness result can be expected. Thus we can only find a syntactic analogue of the consistency of a single formula rather than being able to find an analogue of the general consequence relation. It will be seen that the weak form of completeness is still very useful: in fact the distinction between weak and strong here is subtle enough to be often ignored.

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Then, we present the axiomatization, and comment on its obvious soundness.

It is important to note that like our result in [9] for the reals and unlike that in [3], our axiomatization does not use any unorthodox rules of inference such as the IRR rule of [7]. For reasons discussed in [9] and [12], this rule is slightly controversial and it is important to show that it is not necessary.

In most of the rest of the paper we prove the completeness result devoting sections to finding a rational-flowed model, proving expressive completeness, proving Dedekind completeness for a certain type of definable equivalence, finding a integer-flowed model and concluding the proof.

The integer numbers flow of time is obviously a useful model for situations in which we have discrete moments of time but have neither a beginning or an end of time. We see an example of a such a situation in the temporal database work of [2] where the axiom system presented here is modified into an axiom system for reasoning with updates of databases.

2 The Logic

In this paper temporal structures, which we will often just call structures, will be linear. Thus they consist of a domain T , an irreflexive linear order $<$ on T and a valuation h assigning each p of a countable set of atoms to a subset $h(p) \subseteq T$. The underlying linear order $(T, <)$ of a structure $(T, <, h)$ is called the flow of time of $(T, <, h)$.

The temporal language will usually be that generated by the connectives U and S . The set of formulas is defined recursively to contain the atoms, $\top$ and $\perp$ and for formulas A and B , we include $\neg A$ and $A \wedge B$ along with $U(A, B)$ and $S(A, B)$. $U(A, B)$ is read “until A , B ” or “ B until A ”. Similarly S is to be read as “since”.

Formulas are evaluated at points of the flow of time. The readings above suggest the semantics but more formally truth is defined recursively as follows. Suppose that, inductively, we have defined the truth of formulas A and B in a structure $(T, <, h)$ at all points t : now define

$$\begin{aligned}
 (T, <, h) \models p(t) & \quad \text{iff } t \in h(p), \text{ for } p \text{ atomic} \\
 (T, <, h) \models \top(t) & \\
 (T, <, h) \not\models \perp(t) & \\
 (T, <, h) \models (\neg A)(t) & \quad \text{iff } (T, <, h) \not\models A(t) \\
 (T, <, h) \models (A \wedge B)(t) & \quad \text{iff } (T, <, h) \models A(t) \text{ and } (T, <, h) \models B(t) \\
 (T, <, h) \models U(A, B)(t) & \quad \text{iff there is } s > t \text{ such that} \\
 & \quad (T, <, h) \models A(s) \text{ and} \\
 & \quad \text{for all } u \in T, \text{ if } t < u < s \text{ then } (T, <, h) \models B(u) \\
 (T, <, h) \models S(A, B)(t) & \quad \text{iff there is } s < t \text{ such that} \\
 & \quad (T, <, h) \models A(s) \text{ and} \\
 & \quad \text{for all } u \in T, \text{ if } s < u < t \text{ then } (T, <, h) \models B(u)
 \end{aligned}$$

Often, because of the symmetry of their definitions, results involving U and/or S have *dual* or *mirror* versions which can be stated and proven by simply

swapping U and S for each other and swapping $<$ and $>$ in the original. We mention these frequently but, of course, never bother to prove them.

As usual we have all sorts of abbreviations: the classical ones $\vee$, $\rightarrow$ and $\leftrightarrow$ along with

- FA for $U(A, \top)$ - A will be true,
- PA for $S(A, \top)$ - A was true,
- GA for $\neg F\neg A$ - A will always be true,
- HA for $\neg P\neg A$ - A was always true,
- K^+A for $\neg U(\top, \neg A)$ - A will be true arbitrarily soon and
- K^-A for $\neg S(\top, \neg A)$ - A was true arbitrarily recently.

For formula A and set Γ of formulas we write $\Gamma \models A$ iff for all valuations h into $\mathbb{Z}$, for all $t \in \mathbb{Z}$, if for all $B \in \Gamma$, $(\mathbb{Z}, <, h) \models B(t)$ then $(\mathbb{Z}, <, h) \models A(t)$. We are trying to find a syntactic equivalent to this consequence relation.

In the proof we will have to consider flows of time other than integers but throughout this paper all temporal structures will be assumed to have linear flows.

3 Weak versus Strong Completeness

We are only going to be able to prove a weak completeness result for this logic. Let us review the concepts involved before seeing why. Suppose that we are dealing with a class $\mathcal{K}$ of flows of time and an axiom system Z for the logic of some temporal language $\mathcal{L}$ over $\mathcal{K}$. Assume the usual definition of a *proof* (in Z) of a formula A from a set Γ of formulas. Write $\Gamma \vdash_Z A$ if there is such a proof. Since each inference rule only has one formula as a conclusion in such axiom systems we will call them *finitary* axiom systems.

Assume all the usual definitions for

- syntactic concepts like theorems, consistent sets, consistent formulas and maximal consistent sets for Z ,
- semantic concepts like $\models_{\mathcal{K}}$, models of sets of formulas, satisfiability and validity for $\mathcal{K}$ and
- soundness of Z for $\mathcal{K}$.

Recall that Z is *strongly complete* (often just written *complete*) for $\mathcal{K}$ iff one of the following two equivalent conditions hold:

- if Γ is Z -consistent then Γ has a model in $\mathcal{K}$,
- for all Γ , A , if $\Gamma \models_{\mathcal{K}} A$ then $\Gamma \vdash_Z A$.

To show that we can not have a strongly complete axiomatization in the case of U and S over the integers we will use the concept of compactness. We say that a logic has the *compactness property* iff every unsatisfiable set of formulas has an unsatisfiable finite subset.

Theorem 1. *If a logic has a sound and strongly complete finitary axiomatization then it also has the compactness property.*

This follows easily from the definitions as a proof of inconsistency of a set only uses a finite number of formulas from that set.

It follows that there can be no sound and strongly complete finitary axiomatization of the logic of U and S over integer time. This is because we can see that the logic is not compact by considering the set $\{A_n \mid n \in \mathbb{N}\}$ where $A_0 = F(q \wedge H(\neg q))$ and for each $n \in \mathbb{N}$, $A_{n+1} = F(p \wedge A_n)$. The set is clearly unsatisfiable while all its finite subsets are.

Fortunately, there is a weaker notion of completeness which is still useful. We say Z is *weakly complete* for $\mathcal{K}$ iff one of the following two equivalent conditions hold:

- if A is Z -consistent then A has a model in $\mathcal{K}$,
- for all finite Γ , for all A , if $\Gamma \models_{\mathcal{K}} A$ then $\Gamma \vdash_Z A$.

In the rest of this paper we will show that this is indeed a weaker notion by showing that the logic of U and S over the integers does admit a weakly complete axiomatization.

4 The Axiomatization

Our system **US/Z** has the usual rules for a temporal logic: i.e. modus ponens, generalizations and substitution:

$$\frac{A, A \rightarrow B}{B} \quad \frac{A}{GA} \quad \frac{A}{HA} \quad \frac{A(q)}{A(q/B)}$$

The axioms of **US/Z** are:

- all classical tautologies,
- the six Burgess-Xu axioms

$$\begin{aligned} G(p \rightarrow q) &\rightarrow (U(p, r) \rightarrow U(q, r)) \\ G(p \rightarrow q) &\rightarrow (U(r, p) \rightarrow U(r, q)) \\ p \wedge U(q, r) &\rightarrow U(q \wedge S(p, r), r) \\ U(p, q) &\rightarrow U(p, q \wedge U(p, q)) \\ U(q \wedge U(p, q), q) &\rightarrow U(p, q) \\ U(p, q) \wedge U(r, s) &\rightarrow \\ &U(p \wedge r, q \wedge s) \vee U(p \wedge s, q \wedge s) \vee U(q \wedge r, q \wedge s) \end{aligned}$$
 along with each of their duals,
- plus axioms for discreteness and no end points:

$$U(\top, \perp) \text{ and } S(\top, \perp),$$
- and suitable versions of the Prior axioms:

$$\begin{aligned} \text{Prior-UZ: } &Fp \rightarrow U(p, \neg p) \\ \text{Prior-SZ: } &Pp \rightarrow S(p, \neg p) \end{aligned}$$

Soundness is clear: we are going to spend the rest of the time proving *weak* completeness. The next few sections establish a variety of useful preliminary results.

Notions such as maximal consistent and $\vdash$ are assumed to refer to the logic of U and S over the integers and our system **US/Z** unless otherwise specified.

5 The Burgess-Xu Result

Our task in this section is to find a model of a consistent formula which has a vaguely integer-like flow of time and is one in which $U(\top, \perp)$, $S(\top, \perp)$ and all substitution instances of the Prior axioms are valid. Fortunately, most of the work has been done already. Burgess in [1] proves soundness and completeness of a set of axioms for linear time. Xu, in [13], simplifies the set of axioms and the proof.

Theorem 2. *The Burgess-Xu system (the six axioms and duals, propositional tautologies and the four rules) is sound and strongly complete for the US logic on the class of all linear frames.*

Although neither Burgess nor Xu mention *strong* completeness their proofs do establish that. This is just as well for we need strong completeness.

Let us see how we can use this theorem.

Now suppose that we have a set Γ of formulas consistent with the system **US/Z**. Knowing Lindenbaum's lemma, we can, without loss of generality, assume that Γ is maximal consistent.

By the theorem, since Γ is also consistent with the Burgess-Xu system, there will be a linear model for Γ : i.e. a linear structure in which there is a point, t say, at which all the formulas in Γ hold.

By looking at Burgess's construction (or using Löwenheim-Skolem) we can suppose that the structure is countable.

Since $GU(\top, \perp)$ and its mirror must be true at t , the order does not have end points and the order is discrete.

Because it says so in Γ , all the substitution instances of the other axioms hold everywhere so we have...

Corollary 3. *For every **US/Z**-consistent set Γ of formulas, there is a temporal structure M and $t \in M$ such that*

1. *the flow of time of M is countable, discrete and without end points,*
2. *for all $A \in \Gamma$, $M \models A(t)$ and*
3. *all substitution instances of the axioms Prior-UZ and Prior-SZ are valid in M .*

6 Expressive and Dedekind Completeness

An important technique in our proof is that of switching between the temporal language and an associated first-order one. Let us introduce the concepts and results needed.

We will associate a temporal language with a first-order one called the *monadic* language because it is built from a signature containing only 1-ary predicate symbols along with the binary $<$ predicate symbol. Each atom p in

the temporal language corresponds to a predicate symbol P . We can make a temporal structure $(T, <, h)$ into a first-order structure in the monadic language, by interpreting $<$ as $<$ and each P as being true of exactly those points in $h(p)$.

If, as will later be the case, we restrict to a temporal language with a finite number of atoms then the monadic signature is finite but otherwise it will contain a countable number of 1-ary predicate symbols.

It turns out, unsurprisingly, that the temporal formula $U(p, q)$ is true at exactly those points in a structure where the monadic formula

$$\psi_{U(p,q)}(t) = \exists s > t (P(s) \wedge \forall u (t < u \wedge u < s \rightarrow Q(t)))$$

holds. A simple induction, (see for example [5]), establishes that all temporal formulas A have a corresponding monadic formula ψ_A in one free variable such that, for all structures $(T, <)$, for all valuations h , for all $t \in T$,

$$(T, <, h) \models A(t) \text{ iff } (T, <, h) \models \psi_A(t).$$

We call ψ_A the *table* of A . The induction generalises to show that provided the connectives of the language have first-order tables, as U and S do, then all the temporal formulas of any temporal language have first-order tables.

One may ask whether all first-order formulas with one free variable can be got as tables of temporal formulas. This, of course, depends on the temporal connectives used in the language, but it also depends on what class of structures we restrict attention to. Let us be more precise. Suppose that $\mathcal{S}$ is a class of temporal structures. We say that a temporal language is *expressively complete* over $\mathcal{S}$ if and only if for each monadic formula $\phi(t)$ with one free variable, there is a temporal formula A of the temporal language such that for all $(T, <, h) \in \mathcal{S}$, for all $t \in T$,

$$(T, <, h) \models \phi(t) \text{ iff } (T, <, h) \models A(t).$$

Note the uniformity of the translation over the whole of $\mathcal{S}$.

One of the first expressive completeness results was that of Kamp's in [8]. Kamp showed expressive completeness for (the language with) U and S over the class of all structures whose underlying flow of time is Dedekind complete.

As shown in [5], U and S are still expressively complete even if we allow isolated gaps in the structure but as has been known for while, and as is shown in [5], lemma 3, over the whole class of structures whose underlying flow of time is linear, U and S are not expressively complete. To achieve expressive completeness for the class of all structures with linear flows we need to use the so called Stavi connectives U' and S' which were defined in [4]. $U'(A, B)$ holds if B is true from now until a gap in time after which B is arbitrarily soon false but after which A is true for a while: $U'(A, B)$ is as pictured

$$\begin{array}{c} B \quad \leq \cdots \neg B \\ \hline \text{now} \quad \text{a gap} \quad A \end{array}$$

S' is defined dually. Despite involving a gap, U' is in fact a first-order connective and its table is given by:

$$\begin{aligned}
U'(p, q) \equiv & \\
\exists s \quad t < s \quad & \\
\wedge \forall u \quad (& \quad t < u < s \rightarrow \\
& ([\quad \exists v(u < v \wedge \forall w(t < w < v \rightarrow q(w)) \quad] \\
& \vee [\quad \forall v(u < v < s \rightarrow p(v)) \\
& \quad \wedge \quad \exists v(t < v < u \wedge \neg q(v)) \quad])) \\
& \wedge \exists u[t < u < s \wedge \neg q(u)] \\
& \wedge \exists u[t < u < s \wedge \forall v(t < v < u \rightarrow q(v))]
\end{aligned}$$

We have

Theorem 4. *The language with $\{U, S, U', S'\}$ is expressively complete for the class of structures with linear flow of time.*

This result is mentioned in [4] without proof. The first published proof - a direct proof - is in [5]. In [6] is a proof using the separation technique of Gabbay.

Obviously there is some connection between definable gaps and our Prior axioms. Call a linear temporal structure a *Prior structure* if it satisfies all substitution instances of

$$\text{Prior-U: } U(\top, p) \wedge F\neg p \rightarrow U(\neg p \vee K^+(\neg p), p)$$

and its dual Prior-S. It is easy to see that then there are no definable gaps. Note that this result also holds for our stronger Prior axioms Prior-UZ and Prior-SZ (the weaker axioms are useful in non-discrete structures). It is now not hard to prove the following (see [3], proposition 4.2).

Theorem 5. *The language with U and S is expressively complete for the class of Prior structures.*

Proof. By the expressive completeness of $\{U, S, U', S'\}$ over all linear structures, it suffices to prove that for any $\{U, S, U', S'\}$ -formula B' , there is a $\{U, S\}$ -formula B such that $B' \leftrightarrow B$ is valid in all Prior structures.

This can be achieved by a simple induction on the construction of such B' . The cases of atoms and $\wedge, \neg, U$ and S are immediate. Let us look at $U'(A, B)$ when, by induction, we can suppose that A and B are US -formulas. We claim that $U'(A, B) \leftrightarrow \perp$ is valid in all Prior structures.

Suppose for contradiction that $M \models U'(A, B)(t)$ in some Prior structure M . Thus B holds for a while up until a gap after which $\neg B$ is true arbitrarily soon. By Prior-U applied to B we have $M \models U(\neg B \vee K^+(\neg B), B)(t)$ which is the contradiction.

The case of S' is similar.

7 No gaps between equivalence classes

We know that the Prior axioms ensure that there will not be any definable gaps in a model. To show that our model can be made into a model over the integers

we actually need a stronger result. We need to know that a certain type of definable equivalence relation also does not have its equivalence classes ending at gaps. First some definitions.

The *intervals* of a structure M are just the convex subsets of M and we will use the usual $(a, b]$, etc., notation for them.

If S is a subset of the domain of a temporal structure M (usually S will be an interval here) then we write $M \upharpoonright S$ for the temporal structure with domain S , ordering as in M and interpretation of atoms just the restrictions of the interpretations in M to S . $M \upharpoonright S$ is called the *substructure* of M with domain S .

Suppose that $\varepsilon(x, y)$ is a monadic formula with two free variables x and y . We say that ε defines a *contemporaneous* equivalence relation if and only if on any temporal structure M , if we define the binary relation $\sim_M$ by

$$a \sim_M b \text{ iff } \models \varepsilon(a, b),$$

then

- $\sim_M$ is an equivalence relation on the domain of M ,
- $\sim_M$ partitions M into intervals and
- ε depends only on contemporary properties: i.e. for all $a, b \in M$,

$$\models \varepsilon(a, b) \text{ iff } \models [a, b] \models \varepsilon(a, b).$$

A binary relation $\sim$ on a structure M is called a *contemporaneous* equivalence relation if and only if it is defined as $\sim_M$ by such an ε .

We prove that no so defined contemporaneous equivalence relation has equivalence classes ending at gaps in any Prior structures.

Given such an ε , define $\rho(x)$ as

$$\begin{aligned} & \exists y > x \neg \varepsilon(x, y) \\ & \wedge \neg \exists z (\varepsilon(x, z) \wedge \forall y > z \neg \varepsilon(x, y)) \\ & \wedge \neg \exists z (x < z \wedge \neg \varepsilon(x, z) \wedge \forall y (x < y < z \rightarrow \varepsilon(x, y))). \end{aligned}$$

This says that x 's $\sim$ -class ends in a gap on the right. Dually we can define $\lambda(x)$ about left ends. Note that the end of the whole structure is not a gap and that ρ will not hold of points in the last $\sim_M$ class (if there is such a class).

Now by the expressive completeness of U and S there is temporal R true in any Prior structure exactly where $\rho(x)$ is.

Lemma 6. *Suppose that ε defines the contemporaneous equivalence relation $\sim_N$ on any structure N .*

Then there is an US -formula R which holds in any Prior structure N exactly at those points whose $\sim_N$ -class ends in a gap on the right.

Dually L .

Now suppose that M is a Prior structure and that $\sim = \sim_M$ is a contemporaneous equivalence relation defined by ε .

Lemma 7. *The maximal intervals in which R holds are open intervals which, if bounded, have elements of M as their (excluded) end points.*

Proof. Suppose that R holds at $t \in M$. Clearly ρ holding at t implies that R will hold for a while after t : up until a gap in fact. Thus t is in a non-singleton interval of R . It is possible that R holds for ever after t .

If R does not hold for ever after t then Prior-U applied to R implies that M contains a last point of this stretch of R (plainly impossible given ρ) or a first point of $\neg R$. This is as claimed.

Now look to the left of t . Looking back from just after t we can use Prior-S and see that either R is true always before t , there is a last point of $\neg R$ just before this stretch of R or there is no last point of $\neg R$, but instead a first point of R . We must rule out the third case. Note that in the case of M not being dense there may be both a last point of $\neg R$ and a first point of R : this possibility, subsumed in the second case above, is acceptable in that it implies an excluded end point.

Suppose, for contradiction, that s is this first point of R so that $M \models (R \wedge K^-(\neg R))(s)$. The $\sim$ -class containing s can not stretch for ever into the future for then it does not end in a gap. Neither can it stretch to the end of the maximal interval of R as it would again not end at a gap.

Thus there are other classes in this interval continuing on the other side of the gap which ends s 's. And for a while after the gap R continues to be true: we have not reached the end of the interval yet. Thus $R \wedge K^-(\neg R)$ does not hold at the left hand end of any of these classes.

Let B be the temporal formula saying that the $\sim$ -class we are now in begins with a point satisfying $R \wedge K^-(\neg R)$. B exists by expressive completeness. B holds in s 's class up to the gap and is false arbitrarily soon after the gap. This contradicts Prior-U applied to B .

Lemma 8. *There is no last class and no first class in any maximal interval of R .*

Proof. The last class in a maximal interval of R wouldn't end in a gap.

By expressive completeness, the formula

$$\rho(x) \wedge \forall y < x (\neg \varepsilon(x, y) \rightarrow \exists z (y < z < x \wedge \neg \rho(z)))$$

has a temporal equivalent which is true only in the first classes of maximal intervals of R . If there is a first class then no immediately subsequent classes satisfy this and so we have this formula holding up to a gap and false arbitrarily soon afterwards. This contradicts Prior-U.

Lemma 9. *If a temporal formula holds somewhere in one $\sim$ -class in a maximal interval of R , then it holds somewhere in each $\sim$ -class in the interval.*

Furthermore, each pair of the $\sim$ -classes in a maximal interval of R are elementarily equivalent (taken as substructures of M).

Proof. For a contradiction to the first statement, suppose that A holds in one class but not anywhere in some other class in the same maximal interval of R .

Using expressive completeness and ε , find B which is true at points only if A occurs somewhere in their $\sim$ -class. By using $\neg B$ instead if necessary we may suppose that we have B holding throughout one $\sim$ -class in our maximal interval of R and false throughout a later class. Choose a point t in this former class in which B holds. B holds in the whole of a class if it is true anywhere at all in the class so it continues for a while after t . By Prior-U there is either a last point where B holds after t (not possible as B must continue for a while) or a first point $s > t$ where $\neg B \wedge K^-(B)$ holds.

So s must be the left hand end point of its $\sim$ -class. Look at the gap at right hand end of this class. We can not have B arbitrarily soon after the gap because of Prior-U. Thus for a while after this class B stays false.

Let C be the temporal formula saying that we are now in a class whose left hand end point is also in the class and at that point $K^-(B)$ holds. Now C is true in s 's class but false afterwards contradicting Prior-U.

Now consider the second statement in the lemma. Given a monadic sentence ϕ , we relativise it by restricting quantifiers to where $\varepsilon(x, -)$ holds. We get a formula $\phi(x)$ of one free variable. By expressive completeness this is equivalent to a temporal formula. This is true exactly throughout $\sim$ -classes which model ϕ . Then, by the first part of the lemma, it can't be true somewhere and false elsewhere in the interval.

We define a *bad* point to be where $R \vee L$ holds. We define a *bad* interval as a non-empty and maximal one in which $R \vee L$ holds throughout.

Lemma 10. *Bad points only occur in non-singleton bad intervals.*

In any bad interval both R and L hold throughout. Any bad interval, if bounded, has excluded end points in M (neither R nor L holds at these end points).

Proof. We first show that L holds wherever R does. Suppose for contradiction that we have a maximal interval of R in which L fails to hold somewhere. So $\neg L$ holds throughout at least one $\sim$ -class. By the definition of L , there are two cases. Either this particular $\sim$ -class is one which includes its left hand end point or it is one which begins just after some point of M . The class can not be unbounded below for then it would be first in this bad interval.

In fact we can not have a class beginning just after a point r of M . Since the class can not be first in the bad interval r itself must be in a $\sim$ -class in the bad interval. But r 's class can not end in a gap on the right when r must be its right hand end point.

Thus we have a class in the bad interval which includes its left hand end point. Its not hard to use the previous result to show that throughout the bad interval all classes include their left hand end points.

Let B be a temporal formula true at times which are not left hand end points of their $\sim$ -classes. B is then true continuously in any class from just after the

left hand end point up until the gap at the right hand end point. B must be false arbitrarily soon after the gap contradicting Prior-U.

Using mirror images of the above and previous results we get our proof.

Lemma 11. *If a formula B is true for a while at the start of a $\sim$ -class in a bad interval then it holds throughout the bad interval. Similarly at the end.*

If a formula is true anywhere in a bad interval it is true arbitrarily close to each end of each class in the interval.

Proof. Suppose that $\gamma < \delta$ are gaps and that (γ, δ) is a $\sim$ -class within a bad interval.

Suppose that B holds for a while after γ but that $\neg B$ holds somewhere in the bad interval. By lemma 4, $\neg B$ also holds somewhere in (γ, δ) .

Using ε and expressive completeness we can find a temporal formula C which is true only at points within a $\sim$ -class after some $\neg B$ in that class. C will be false for a while at the beginning of each class and then true for a while at the end.

In fact C is true for a while up to the gap at the end and false arbitrarily soon after the gap. This contradicts Prior-U.

Applying the above to the negation of a formula gives us the second part.

Let us see what happens if we interfere with M by replacing a whole bad interval by one of its $\sim$ -classes.

Let Q^- be the subset of the domain of M being all that precedes the bad interval. Let Q^+ be all that follows. Either or both of these may be empty. Let Q_0 be the bad interval itself and I be any one of its $\sim$ -classes.

We look at N , the substructure of M whose domain is just $Q^- \cup I \cup Q^+$.

Lemma 12. *For all temporal formulas A , for all $t \in N$,*

$$\models A(t) \text{ iff } N \models A(t)$$

Proof. We proceed by induction on the construction of A . The cases of atomic and boolean A are immediate. Now consider $U(A, B)$: $S(A, B)$ is similar.

$(\Rightarrow)$: Consider then when $M \models U(A, B)(t)$ with $t \in N$. Say that $s \in M$, that $t < s$, that $M \models A(s)$ and for all $u \in M$, if $t < u < s$ then $M \models B(u)$.

There are several cases.

1. $t < s \in Q^-$: Apply the induction hypothesis to A and B at s and at all points in between.
2. $t \in Q^-$ and $s \in Q_0$: A holds somewhere in Q_0 so somewhere in I (by lemma 6). So holds there in I in N . B holds for a while into Q_0 so, by lemma 6, holds everywhere in Q_0 . By the induction hypothesis, B holds everywhere in I in N . Hence result.
3. $t \in Q^-$ and $s \in Q^+$: We can deduce that B holds throughout I in both M and N and get the result.
4. $t < s \in I$: Straight forward use of inductive hypothesis.

5. $t \in I$ and s later in Q_0 : Again by lemma 6 we have B true throughout I in M and so in N . Since A is true somewhere in Q_0 in M , lemma 6 tells us that A is true arbitrarily close to the end of I in M and so in N . This gives us our result.
6. $t \in I$ and $s \in Q^+$: B is true throughout I and we have our result.
7. $t < s \in Q^+$: Apply induction hypothesis to A and B at s and at all points in between.

($\Leftarrow$) : Consider then when $N \models U(A, B)(t)$. Say that $t < s$, that $N \models A(s)$ and for all $u \in N$, if $t < u < s$ then $N \models B(u)$.

Again there are several cases:

1. $t < s \in Q^-$: Apply induction hypothesis to A and B at s and at all points in between.
2. $t \in Q^-$ and $s \in I$: B holds from t up until the end of Q^- in both M and N . B holds at the beginning of I in N and so in M . By lemma 6 B holds throughout Q_0 . A holds in I in N and so in M and we have our result.
3. $t \in Q^-$ and $s \in Q^+$: B holds throughout I in N and so in M . Lemma 6 tells us B holds throughout Q_0 in M .
4. $t < s \in I$: Straight forward use of inductive hypothesis.
5. $t \in I$ and $s \in Q^+$: B is true throughout I and we have our result.
6. $t < s \in Q^+$: Apply induction hypothesis to A and B at s and at all points in between.

Lemma 13. *In fact there can't have been any bad points anyway.*

Proof. By lemma 7, R holds in I in N .

But by lemma 1, R holds at a point in any Prior structure (not just M) if and only if the $\sim$ -class of the point ends in a gap (where $\sim$ is the appropriate equivalence relation for the structure). And N is a Prior structure: we still have all the instances of Prior-U/S continuing to hold as any counterexample point in N is also one in M .

By the contemporaneity of ε , I as a subset of N , like I as a subset of M , is all in one $\sim_N$ -class. Could the class be bigger now?

R is true of this class so that it is bounded above amongst other things. Thus Q^+ is non-empty and by lemma 5 begins with a point q say. Also by lemma 5 $\neg R$ holds at q in M and so in N . Clearly q is not in the class of I in N . Thus the class ends just before q .

R can not have been true in this class after all.

Thus we have proven...

Theorem 14. *Suppose that $\sim$ is a contemporaneous equivalence relation on a Prior structure M .*

Then the $\sim$ -classes do not end at gaps.

8 Using Contemporaneity on the Integers

Let us see how we can use the theorem above in our proof. Note that here we are working in a language with only a finite number of atoms though.

Theorem 15. *Suppose that M is a temporal structure in a finite language such that*

- *the flow of time of M is countable, discrete and without end points,*
- *all substitution instances of the axioms Prior-UZ and Prior-SZ are valid in M .*

Then for all $k < \omega$, there is a temporal structure with flow of time the integers satisfying the same monadic first-order sentences of quantifier depth at most k as M does.

Proof. First some preliminaries. Fix $k \geq 3$.

Here a *structure* will mean a linear temporal structure in our finite language.

If M and N are structures we write $M \equiv_k N$ if and only if M and N agree on the truth of monadic sentences of quantifier depth at most k . Note that since $k \geq 3$, if $M \equiv_k N$ then M and N either both have a right(respectively left) hand end point or both do not have a right(resp. left) hand end point. Discreteness is also preserved.

We assume familiarity with lexicographic sums of linear orders and with the fact that $\equiv_k$ is preserved under such sums. See [10], [3] or [9] for details.

If a is an element of a discrete structure M then we write $a - 1$ for its immediate predecessor if it has one and $a + 1$ for its immediate successor, if it has one.

Say that M is *good* if and only if there is some $N \equiv_k M$ such that the flow of time of N is an interval of the integers.

Say that M is *very good* if and only if, for all $t \leq u$ in M , the substructure $M \upharpoonright [t, u]$ is good.

Lemma 16. *If N is countable and very good then it is good.*

Proof. All finite structures are good so suppose that N has countably infinite domain. If N has two end points then it is clearly good. First consider the case when N has a beginning a_0 but no (right hand) end.

Choose $a_i \in N$ for each positive integer i such that $i < j$ implies $a_i < a_j$ and for all $t \in N$, there is j such that $t < a_j$. Since N is very good, $N \upharpoonright [a_i, a_{i+1} - 1]$ is good. For $i = 0, 1, \dots$, take $Z_i \equiv_k N \upharpoonright [a_i, a_{i+1} - 1]$ with a finite interval of $\mathbb{Z}$ as a flow.

Because $\equiv_k$ is preserved under lexicographic sums,

$$N \equiv_k \sum_{i \in \mathbb{N}} (Z_i)$$

the latter having flow isomorphic to a (half) subinterval of $\mathbb{Z}$.

If N has an end but no beginning then the proof is similar. If N has no end points then choose $a_0 \in N$, use the above arguments on $(-\infty, a_0]$ and $[a_0 + 1, +\infty)$, and then use the lexicographic sum result to add appropriate structures together.

Define $\sim_M$ on a temporal structure M by for any $a, b \in M$, $a \sim_M b$ if and only if

- $a = b$,
- $a < b$ and $M \upharpoonright [a, b]$ is very good or
- $b < a$ and $M \upharpoonright [b, a]$ is very good.

Lemma 17. $\sim_M$ is a contemporaneous equivalence relation on the domain of any M .

Proof. Clearly there are only finitely many logically inequivalent maximal consistent conjunctions γ of sentences of quantifier depth $\leq k$. Any structure is a model of just one such γ , so if $N_1 \models \gamma$ then $N_2 \equiv_k N_1$ iff $N_2 \models \gamma$. Only some will be true of good structures- $\{\gamma_1, \dots, \gamma_s\}$ say. N is good iff $N \models \bigvee_{i \leq s} \gamma_i$.

Let $\gamma(z, t)$ be the result of relativizing the quantifiers of $\bigvee_{i \leq s} \gamma_i$ to $[z, t]$, where z and t are new variables.

Then

$$\begin{aligned} \varepsilon(x, y) = & \quad x < y \rightarrow \forall zt(x \leq z \leq t \leq y \rightarrow \gamma(z, t)) \\ & \wedge y < x \rightarrow \forall zt(y \leq z \leq t \leq x \rightarrow \gamma(z, t)) \end{aligned}$$

is a formula defining $\sim_M$.

To show that $\sim$ is contemporaneous, we first show that it is an equivalence relation. The difficult part is transitivity. Suppose that $a < b < c$ are in M and $a \sim_M b$ and $b \sim_M c$. We show that $M \upharpoonright [a, c]$ is very good by showing that if $a \leq t < u \leq c$ then $M \upharpoonright [t, u]$ is good.

If t and u are on the same side of b then this is clear. If $b = t$ or $b = u$ then use a lexicographic sum.

So assume that $a \leq t < b < u \leq c$. First note that since $M \upharpoonright [b, c]$ is very good then it is also good (even if it is not countable). Thus its flow is discrete and there is a successor $b + 1$ of b . Now $M \upharpoonright [t, b]$ and $M \upharpoonright [b + 1, u]$ are both good. Choose $Z_1 \equiv_k M \upharpoonright [t, b]$ and $Z_2 \equiv_k M \upharpoonright [b + 1, u]$ each with flow a subset of $\mathbb{Z}$. Then we know that $M \upharpoonright [t, u] \equiv_k Z_1 + Z_2$ whose flow is isomorphic to an interval of $\mathbb{Z}$ itself.

That the $\sim_M$ classes are intervals follows from the fact that very goodness is inherited by substructures on subintervals.

Contemporaneity then follows from the fact that the definition of $\sim_M$ is in terms of exactly the right substructure.

Now let us turn to the proof of the main theorem.

If M is good then we are done. So suppose not. Thus M is not very good. So there is $a < b \in M$ such that $M \upharpoonright [a, b]$ is not good. Thus $M \upharpoonright [a, b]$ is not very good and we have two disjoint $\sim$ classes.

Now a 's class can not end at a gap on the right (by theorem 5 and the fact that Prior-UZ and dual imply Prior-U and dual) so it must include a point c but not the successor $c + 1$ of c . This can not be because $M \models [c, c + 1]$, like all finite structures is very good and $\sim$ is transitive.

9 Completeness

Finally

Theorem 18. *The system US/Z is sound and weakly complete for the semantics over structures with integers flow.*

Proof. Soundness is straightforward.

To show weak completeness, we suppose that we are given a formula A_0 consistent with US/Z. We will find a model of it with flow of time the integers.

First use Burgess-Xu Corollary 1 to furnish us with a structure M_0 and $t_0 \in M_0$ such that

1. the flow of time of M_0 is countable, discrete and without end points,
2. $M_0 \models A_0(t_0)$ and
3. all substitution instances of the axioms Prior-UZ and Prior-SZ are valid in M_0 .

By ignoring all the atoms which don't appear in A_0 we have a temporal structure M from a finite language. M is still a model of A_0 .

Thus we can apply theorem 6.

Let k be one greater than the quantifier depth of the table $\alpha(t)$ of A_0 . We have a temporal structure $\mathcal{Z}$, with flow of time the integers, satisfying the same monadic sentences of quantifier depth at most k as M does.

Thus $\mathcal{Z}$ like M is a model of $\exists t\alpha(t)$. Say $b \in \mathcal{Z}$ and $\mathcal{Z} \models \alpha(b)$.

We have $\mathcal{Z} \models A_0(b)$ as promised.

Axiomatizing U and S over the natural numbers can be done in a similar manner.

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